



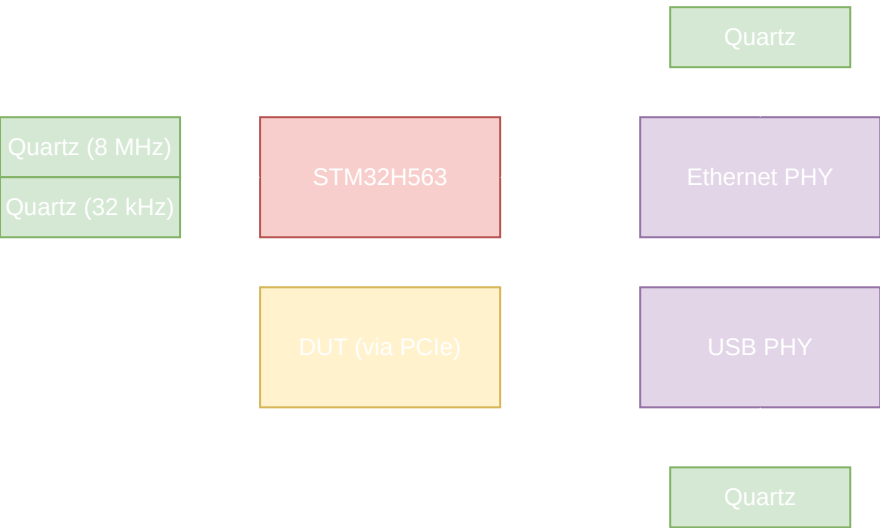
Technical reference manual  
1.0.0

# EtherBench

# Clocking

## Structure

The EtherBench motherboard integrates multiple independent clock domains to ensure optimal signal integrity and flexibility. The primary system clock is derived from an external 8 MHz HSE oscillator.



The STM32H5 uses its internal Main PLL (PLL1) to scale this reference up to a 250 MHz core clock, while its Master Clock Output (MCO1) is routed to the PCIe extension connector to provide a highly configurable clock source for the DUT. Both the Ethernet PHY and USB Hub are autonomous regarding the clocks, as they each have a quartz. The Ethernet PHY provide a clock output, which is used by the Ethernet peripheral.

## Clock outputs

The STM32 provide a clock output, variable for the DUT. There's two choices :

- The core clock, at 250 MHz
- The HSI oscillator, at 64 MHz.

## Available rates

A hardware integer division factor (from 1 to 15) can be applied to either source. While the hardware can physically output high-frequency signals, it is strongly recommended to limit the MCO1 frequency to 64 MHz to preserve signal integrity across the connector interface.

The available output frequencies are mapped below:

| Divisor | HSI64  | PLL     | HSI48  | HSE   | LSE        |
|---------|--------|---------|--------|-------|------------|
| 1       | 32 MHz | 250 MHz | 48 MHz | 8 Mhz | 32.768 kHz |

|    |                  |                  |                 |                 |                   |
|----|------------------|------------------|-----------------|-----------------|-------------------|
| 2  | <b>16 MHz</b>    | <i>125 MHz</i>   | <b>24 MHz</b>   | <b>4 MHz</b>    | <b>16.384 kHz</b> |
| 3  | <i>10.66 MHz</i> | <i>83.33 MHz</i> | <b>16 MHz</b>   | <i>2.66 MHz</i> | <i>10.923 kHz</i> |
| 4  | <b>8 MHz</b>     | <i>62.5 MHz</i>  | <b>12 MHz</b>   | <b>2 MHz</b>    | <b>8.192 kHz</b>  |
| 5  | <b>6.4 MHz</b>   | <b>50 MHz</b>    | <b>9.6 MHz</b>  | <b>1.6 MHz</b>  | <i>6.554 kHz</i>  |
| 6  | <i>5.33 MHz</i>  | <i>41.66 MHz</i> | <b>8 MHz</b>    | <i>1.33 MHz</i> | <i>5.461 kHz</i>  |
| 7  | <i>4.57 MHz</i>  | <i>35.71 MHz</i> | <i>6.85 MHz</i> | <i>1.14 MHz</i> | <i>4.681 kHz</i>  |
| 8  | <b>4 MHz</b>     | <b>31.25 MHz</b> | <b>6 MHz</b>    | <b>1 MHz</b>    | <b>4.096 kHz</b>  |
| 9  | <i>3.55 MHz</i>  | <i>27.77 MHz</i> | <i>5.33 MHz</i> | <i>888 kHz</i>  | <i>3.641 kHz</i>  |
| 10 | <b>6.4 MHz</b>   | <b>25 MHz</b>    | <b>4.8 MHz</b>  | <b>800 kHz</b>  | <i>3.27 kHz</i>   |
| 11 | <i>2.90 MHz</i>  | <i>22.72 MHz</i> | <i>4.36 MHz</i> | <i>727 kHz</i>  | <i>2.979 kHz</i>  |
| 12 | <i>2.66 MHz</i>  | <i>20.83 MHz</i> | <b>4 MHz</b>    | <i>666 kHz</i>  | <i>2.731 kHz</i>  |
| 13 | <i>2.46 MHz</i>  | <i>19.23 MHz</i> | <i>3.69 MHz</i> | <i>610 kHz</i>  | <i>2.521 kHz</i>  |
| 14 | <i>2.28 MHz</i>  | <i>17.85 MHz</i> | <i>3.42 MHz</i> | <i>570 kHz</i>  | <i>2.341 kHz</i>  |
| 15 | <i>2.13 MHz</i>  | <i>16.66 MHz</i> | <b>3.2 MHz</b>  | <i>533 kHz</i>  | <i>2.185 kHz</i>  |

#### Rate selection

The rates in **bold** shall be preferred over the others, as they're integer values, and therefore, more accurates.

## Selection Behavior

When a specific frequency is requested via the CLI or API, the configuration algorithm automatically biases its choice toward these preferred bold values by expanding their error tolerance window to three times that of the non-standard (italic) rates.

In any cases, the selected value **will** be one of the table. Physically, it's not possible to generate any other rate, from hardware limitations. Therefore, for very specific request, the command can introduce an error. The command will return the applied frequency.